



JETHRO MOLEÑO

SOFTWARE ENGINEER

PROFESSIONAL SUMMARY

Entry-level Software Engineer with hands-on experience developing backend services, API-driven applications, and ML-powered systems. Skilled in Python, software debugging, structured QA testing, and building data-driven automation solutions. Demonstrated capability in designing functional software, optimizing model performance, integrating hardware-software systems, and producing reliable end-to-end tools.

PROJECT EXPERIENCE

Automated Fiber Grading System using YOLO and Raspberry Pi 4B

Thesis / Article 1

- Achieved 100% accuracy and 99.33% mAP on 1,443 augmented images from 500 real samples.
- Built full detection pipeline enabling 2–3 second inference time on Raspberry Pi.
- Conducted dataset preparation, augmentation, hyperparameter tuning, and model evaluation.
- Deployed optimized model to edge hardware for automated grading workflow

CreativeCard - Image Generation App using Stable Diffusion XL

OJT Project

- Implemented high-resolution image generation using SDXL 1.0.
- Fine-tuned model for 10 hours achieving 0.0695 MSE loss with ~14.53 seconds per iteration.
- Generated 20+ test outputs and developed CustomTkinter UI for inference.
- Integrated prompt-conditioning and resolution handling for 1500x2000 px outputs.

Tomato Leaf Disease Detection (YOLO)

Academic Project (Intro to AI)

- Developed YOLO-based CV pipeline to detect and classify tomato leaf diseases.
- Executed dataset annotation, preprocessing, augmentation, and model training workflows.
- Tuned model parameters to reduce false positives and improve classification robustness

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Las Piñas City

🌐 [LinkedIn](#) | [GitHub](#)

EDUCATION

FEB 2026

MAPUA UNIVERSITY

- BS Computer Engineering
Specialized in Embedded
Systems

TECHNICAL SKILLS

- Programming: Python, MATLAB, C++, C#, Javascript (basic)
- Software Engineering: REST APIs, modular design, debugging, unit testing, Git/GitHub, SQL (PostgreSQL), Agile/Scrum
- Machine Learning/Computer Vision: YOLOv5/YOLOv8, PyTorch, Tensorflow, OpenCV
- Tools & Frameworks: VS Code, PyCharm, Postman, Docker (basic), Django (basic), Supabase (basic)
- Edge AI / Embedded: Raspberry Pi 4B, ESP8266, sensor integration.
- Other Skills: Technical Documentation, UI/UX Validation

UV Disinfection Robot - Automated & Remote Controlled

OJT Project

- Designed automated movement and remote override logic for UV safety system.
- Integrated hardware controls and embedded logic for robot operation.

Farm Monitoring System Using ESP8266

Academic Project (Embedded Systems)

- Developed IoT-based sensor monitoring with wireless data transmission.
- Implemented threshold-based alerts (ThingSpeak) for environmental conditions.

WORK EXPERIENCE

July 2024 - August 2024

Ollopa Corporation

OJT Trainee

- Conducted structured QA testing on UI components including navigation, inputs, and data views.
- Documented 10+ bugs and produced detailed issue reports for dev team review.
- Supported digital content management for ML-generated card products.

September 2022 - March 2023

iCarCover Inc.

Virtual Assistant

- Managed e-commerce operations and task coordination.

REFERENCES

Engr. Rosemarie V. Pellegrino
Thesis Adviser / MAPUA University
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TRAININGS & SEMINARS

- Deep Learning for Business (Coursera)
- Raspberry Pi & Python Programming (Coursera)
- Software Quality Assurance (CITCO)
- DevOps Fundamentals (Accenture)
- CCNAV7 (Introduction to Networks: Security & Automation)
- Application Development Heroes using Low-Code Platforms (Appia Philippines Inc. / OutSystems)
- Creating Critical “Headless” Applications Using Robotic Process Automation (Appia Philippines Inc.)
- Business Continuity with 3DEXPERIENCE Platform: Collaboration & Project Management (CCESC - MAPUA University)
- Marketing ME (AEYGDS, VXi, Cognizant)

CERTIFICATIONS

- EF SET English Certificate - C2 Proficient

LANGUAGES

- English
- Filipino